An Introduction to AI Agents

What is an AI agent?

An AI agent is a system in which a language model decides which actions to take to reach a goal. Unlike a single prompt-and-response call, an agent runs in a loop: it observes its context, chooses an action such as calling a tool, receives the result, and repeats until it can produce a final answer. The model supplies the reasoning; the surrounding program supplies tools, memory and guard rails.

Planning and task decomposition

Complex questions are usually broken into smaller sub-tasks before any tool is used. A planning step asks the model to list the sub-tasks, the order in which to tackle them and the tool best suited to each. Planning reduces wasted tool calls and makes the agent's behaviour inspectable, because the plan can be logged and reviewed.

Tool use

Tools are described to the model by a name, a description and an input schema. When the model decides a tool is needed it emits a structured tool call; the application executes it and returns the result as an observation. Good tool descriptions matter as much as good code, because the description is how the model learns when to use a tool.

Failure modes

Typical failures include choosing the wrong tool, looping without progress, trusting low-quality search results, and inventing facts when evidence is missing. Mitigations are step limits, forcing a final answer after a maximum number of rounds, requiring citations for every claim, and evaluating outputs automatically.

Memory and retrieval

Agents combine short-term memory, the conversation history, with long-term memory in a vector database. Documents are split into overlapping chunks, embedded, and retrieved by semantic similarity. Retrieval-augmented generation grounds answers in the retrieved passages, which reduces hallucination when combined with citation requirements.